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(54) Title: SMART BINS USING IOT SENSORS

(57) Abstract: The prevent invention is related to the smart bins; which includes various sensors to categorized the waste and the material. The invention is a low-cost method in comparison to the costly and time consuming methods for storing and segregating the waste. The invention is the automated process of doing the waste segregation based on the input sensors and the records are kept on cloud via IOT. With the sensors, the collection and segregation is to be done; where as the transport and storage is for the same type of waste which makes the disposal and re-treatment process easier for the municipal corporation. In addition to the above, system and method also suitable for the hospitals where non-hazardous and hazardous waste are available; the special process of defined in terms of Black bins and Red bins. The detailed description is shown in Figure 1 and Figure 2.

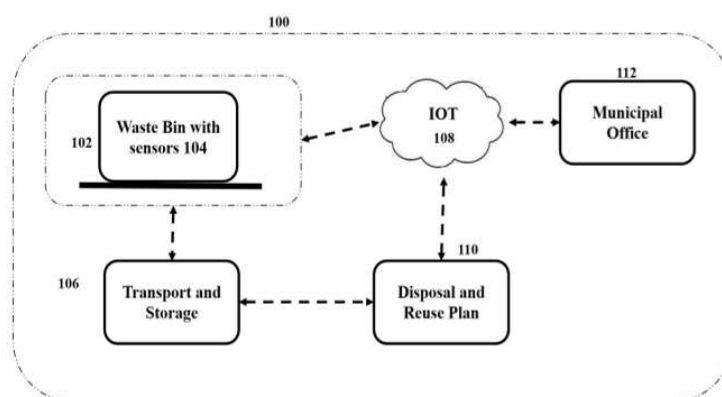


Fig.1

FORM 2

THE PATENTS ACT, 1970

(39 of 1970)

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THE PATENTS RULES, 2003

COMPLETE SPECIFICATION

1. TITLE OF THE INVENTION

SMART BINS USING IOT SENSORS

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3. PREAMBLE TO DESCRIPTION

COMPLETE SPECIFICATION-

The following specification particularly describes the invention and the manner in which it is to be performed.

SMART BINS USING IOT SENSORS

Field of the Invention

[0001] The present invention is the collaborative work of Computer Science with Electronics Engineering; where the sensors and the IOT support is required, the collect and disseminate the data.

Background

[0002] The background description includes information that may be useful in understanding the present invention. It is not an admission that any of the information provided herein is prior art or relevant to the presently claimed invention, or that any publication specifically or implicitly referenced is prior art.

[0003] The present invention aims to make an intelligent waste bin which includes various functionalities discussed in the enclosure.

[0004] US10994928B2 is related to waste receptacles and more particularly to smart waste receptacles capable of identifying waste. A waste receptacle may identify waste using machine vision and perform a task in response to identifying the waste. In mentioned embodiment, the smart waste receptacle may instruct a user to direct the waste to one of a plurality of waste bins based on a category of the waste. In discussed embodiment, the smart waste receptacle may place an order to replenish the stock of a product based on the identification of waste; the smart waste

receptacle is created by attaching a smart device to a standard waste receptacle.

[0005] CN109264259A is related to a kind of dustbins, specially intelligent waste categorised collection dustbin and its categorised collection method, including dustbin upper box body and dustbin lower box, the dustbin cover board that setting can be overturn at the top of dustbin upper box body, the dustbin cover board is equipped with electronic lock ; Marking mechanism is equipped on the inside of dustbin upper box body, dustbin upper box body side wall is additionally provided with side displacement sensor, dustbin upper box body bottom is mechanics tablet, the upper surface of mechanics tablet is equipped with resistance-strain type gravity sensor, it is respectively arranged below with a bottom displacement sensor, the mechanics tablet is openable and closable plate, realizes connection or partition by the folding of mechanics tablet between dustbin upper box body and dustbin lower box, and the dustbin lower box is equipped with garbage outlet.

[0006] In the patent named "Novel and multifunctional intelligent dustbin and application method", discloses novel and multifunctional intelligent dustbin, garbage classification system, more scale interacted systems and anti-pinch hand system are installed on dustbin ; The garbage classification system includes account identification device, host, touch screen and electronic driver, and account identification device includes at least barcode

scanning area, credit card area and recognition of face camera, and barcode scanning area, credit card area and recognition of face camera are connect with host respectively ; The electronic driver is connect with touch screen ; More scale interacted systems include the strain gauge for being respectively arranged at each sorting rubbish cabinet bottom, and each strain gauge is connect by data processor with host ; Anti-pinch hand system includes being set to each inductor classified and thrown above mouth, and inductor signal access host, host is connect with each electrically operated gate electronic driver respectively.

[0007] In another patent, WO2021007875A1, named "Smart dustbin and management system therefor", comprised of four types of bins respectively for recycling and storing plastic trash, metal trash, paper products and organic detritus. The inventors discussed the usage and the description of these bins.

[0008] All publications herein are incorporated by reference to the same extent as if each individual publication or patent application were specifically and individually indicated to be incorporated by reference. Where a definition or use of a term in an incorporated reference is inconsistent or contrary to the definition of that term provided herein, the definition of that term provided herein applies and the definition of that term in the reference does not apply.

[0009] In some embodiments, the numbers expressing quantities of

ingredients, properties such as concentration, reaction conditions, and so forth, used to describe and claim certain embodiments of the invention are to be understood as being modified in some instances by the term "about." Accordingly, in some embodiments, the numerical parameters set forth in the written description and attached claims are approximations that can vary depending upon the desired properties sought to be obtained by a particular embodiment. In some embodiments, the numerical parameters should be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Notwithstanding that the numerical ranges and parameters setting forth the broad scope of some embodiments of the invention are approximations, the numerical values set forth in the specific examples are reported as precisely as practicable. The numerical values presented in some embodiments of the invention may contain certain errors necessarily resulting from the standard deviation found in their respective testing measurements.

[0010] As used in the description herein and throughout the claims that follow, the meaning of "a," "an," and "the" includes plural reference unless the context clearly dictates otherwise. Also, as used in the description herein, the meaning of "in" includes "in" and "on" unless the context clearly dictates otherwise.

[0011] The recitation of ranges of values herein is merely intended to serve as a shorthand method of referring individually to each separate value falling within the range. Unless otherwise indicated herein, each individual

value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g. "such as") provided with respect to certain embodiments herein is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention otherwise claimed. No language in the specification should be construed as indicating any non-claimed element essential to the practice of the invention.

[0012] Groupings of alternative elements or embodiments of the invention disclosed herein are not to be construed as limitations. Each group member can be referred to and claimed individually or in any combination with other members of the group or other elements found herein. One or more members of a group can be included in, or deleted from, a group for reasons of convenience and/or patentability. When any such inclusion or deletion occurs, the specification is herein deemed to contain the group as modified thus fulfilling the written description of all Markush groups used in the appended claims.

Objects of the Invention

[0013] India is confronted with enormous environmental issues related to garbage creation and insufficient trash collection, transportation, treatment, and disposal.

[0014] The Internet of Things (IOT) should be able to integrate a large number of various and heterogeneous end systems in a transparent and seamless manner, while also allowing open access to specified subsets of data for the creation of digital services.

[0015] Environmental waste has a considerably greater impact on society's health. One of the most pressing issues of our day is the identification, monitoring, and management of trash. So, The possible solution for the problem of waste of bins is using IoT sensor based bins. The main features of the IoT based bins would be –

- Level Sensor
- PIR Sensor
- Plastic Sensor
- Metal Sensor
- Temperature Sensor

[0016] Currently, available bins in the market are costly when compared with their uses for waste management. The proposal model will be suitable for smart waste management which would help industry and society. This would act as a building blocks for smart cities.

[0017] With the present invention, the additional feature is to make bins for hospitals: Black bin and Red bin.

Black bins:

- For non-infectious waste i.e. paper waste, food packing , wrappers etc.

Red Bins:

For infectious waste (clinical laboratory waste, blood bank waste, syringes, sharp objects).

[0018] Main objective of the invention is Content Analysis. The plastic material present in the bins and the metal present in the bins is analyzed. According to the maximum presence of the material will decide to which industries the bins material will be displaced.

[0019] The wastes are categorized as:

Non- Hazardous Waste:

- 85% of generated wastes

- Wastes constituting residue of food and peels of fruits, packaging materials(papers & cardboards)

Hazardous Waste:

- Infectious waste
- Medical and biomedical waste
- Contaminated waste

Drawings

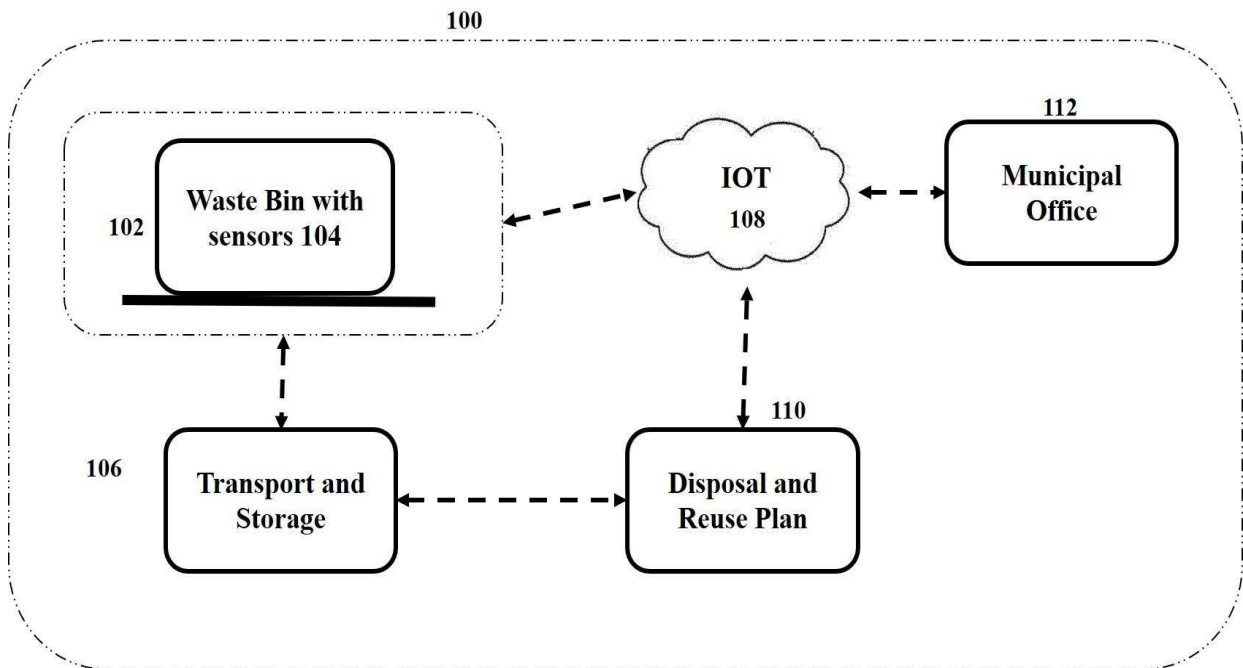


Fig.1

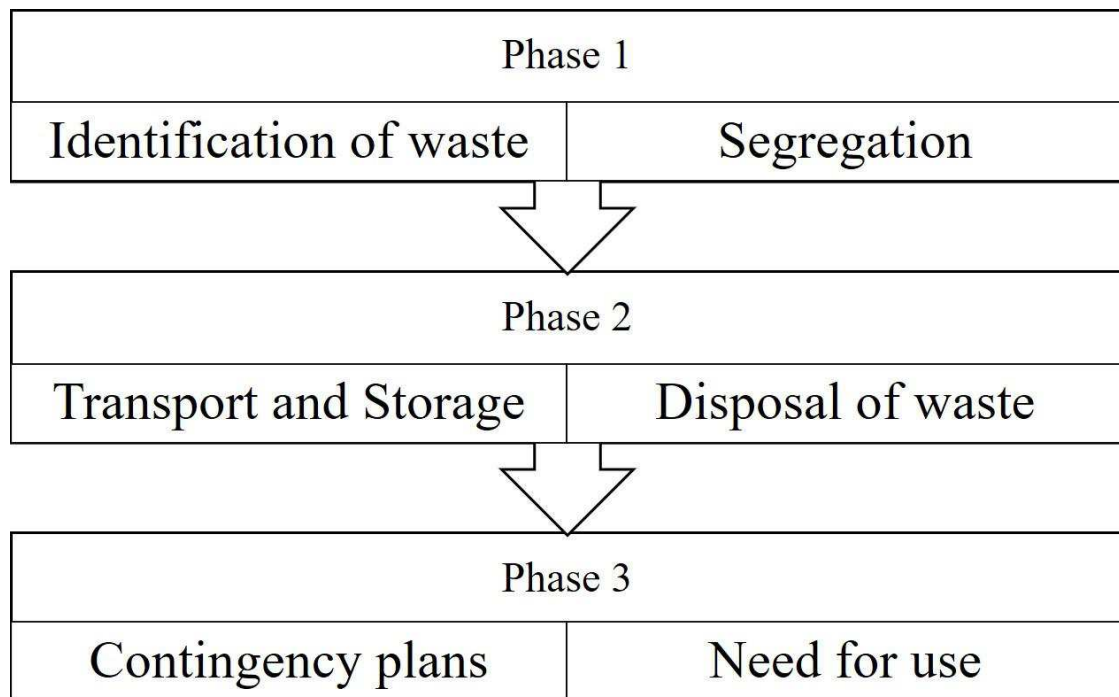


Fig. 2

Brief Description of the Drawings

[0020] FIG. 1 illustrates an exemplary architecture of smart bins, in accordance with an embodiment of the present disclosure.

[0021] FIG. 2 describes the process in three phases related to the enclosure.

Detailed Description

[0022] In Figure 1, The present invention 100, relates to the architecture of the model for smart bins where smart bins 102 is place and the segregation of waste is done based on sensors 104.

[0023] Once the waste is collected, the next step is to transport and store it in appropriate dump yard 106.

[0024] Proper disposal of waste and the re-treatment process is the important step 110 of the present invention.

[0025] All the information about the waste and the smart bins are gathered and monitored via IOT 108; and informed to municipal office 112, for records and analysis purpose.

[0026] The system not only use for the local societies and roads; but also been used at hospitals; where two smart bins are defined: Black bin and Red Bin.

[0027] The process flow is shown in Figure 2 to understanding working of the present invention.

[0028] The following discussion provides many example embodiments of the inventive subject matter. Although each embodiment represents a single combination of inventive elements, the inventive subject matter is considered to include all possible combinations of the disclosed elements. Thus, if one embodiment

comprises elements A, B, and C, and a second embodiment comprises elements B and D, then the inventive subject matter is also considered to include other remaining combinations of A, B, C, or D, even if not explicitly disclosed.

[0029] The term “non-transitory storage device” or “storage” or “memory,” as used herein relates to a random-access memory, read only memory and variants thereof, in which a computer can store data or software for any duration.

[0030] It should be apparent to those skilled in the art that many more modifications besides those already described are possible without departing from the inventive concepts herein. The inventive subject matter, therefore, is not to be restricted except in the spirit of the appended claims. Moreover, in interpreting both the specification and the claims, all terms should be interpreted in the broadest possible manner consistent with the context. In particular, the terms “comprises” and “comprising” should be interpreted as referring to elements, components, or steps in a non-exclusive manner, indicating that the referenced elements, components, or steps may be present, or utilized, or combined with other elements, components, or steps that are not expressly referenced. Where the specification claims refer to at least one of something selected from the group consisting of A, B, C and N, the text should be interpreted as

requiring only one element from the group, not A plus N, or B plus N, etc.

[0031] The following discussion provides many example embodiments of the inventive subject matter. Although each embodiment represents a single combination of inventive elements, the inventive subject matter is considered to include all possible combinations of the disclosed elements. Thus, if one embodiment comprises elements A, B, and C, and a second embodiment comprises elements B and D, then the inventive subject matter is also considered to include other remaining combinations of A, B, C, or D, even if not explicitly disclosed.

Advantages of the Invention

[0032] An advantage of the present invention is to make the system easier for the local bodies and to reduce the man power.

[0033] Currently, available bins in the market are costly when compared with their uses for waste management. The proposal model will be suitable for smart waste management which would help industry and society. This would act as a building blocks for smart cities.

[0034] Content Analysis is the major advantage of the invention; which analyze the presence of plastic material and the metal presence in the bins. According to the maximum presence of the material will decide to which industries the bins material will be displaced.

[0035] Special arrangement for hospitals and the bio-waste.

Claims

We Claim:

Claim 1. The present invention focuses on the smart bins, which embodies the collaboration of electronics engineering field where sensors are used with the mechanical motors to work with the IOT support.

Claim 2. In the system, Five sensors are used:

- Level Sensor
- PIR Sensor
- Plastic Sensor
- Metal Sensor
- Temperature Sensor

Claim 3. The system in claim 2, tends the collection and segregation of the waste based on the material; sensors will segregate the waste.

Claim 4. The contribution of the present system claims:

- i. Content Analysis.
- ii. Transport and storage
- iii. Disposal and re-use treatment

Abstract

The present invention is related to the smart bins; which includes various sensors to categorize the waste and the material. The invention is a low-cost method in comparison to the costly and time-consuming methods for storing and segregating the waste. The invention is the automated process of doing the waste segregation based on the input sensors and the records are kept on cloud via IOT.

With the sensors, the collection and segregation is to be done; whereas the transport and storage is for the same type of waste which makes the disposal and re-treatment process easier for the municipal corporation. In addition to the above, system and method also suitable for the hospitals where non-hazardous and hazardous waste are available; the special process is defined in terms of Black bins and Red bins. The detailed description is shown in Figure 1 and Figure 2.

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